Comprehensive Study Notes: Natural Language Processing & Text Mining

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1. Introduction to Natural Language Processing

Natural Language Processing is defined as the interdisciplinary field that gives computers the ability to understand, interpret, and manipulate human language. The foundational origins of computational linguistics date back to 1950 when Alan Turing proposed the Turing Test in the United Kingdom to evaluate machine intelligence. Modern NLP bridges traditional linguistics, statistical machine learning, and deep neural architectures.

In 1957, Noam Chomsky published Syntactic Structures at MIT in Massachusetts, establishing transformational generative grammar. Contemporary NLP systems developed by Google, IBM, and OpenAI rely heavily on massive textual corpora and distributional representations.

2. Text Preprocessing Pipelines

Tokenization refers to the process of segmenting a continuous stream of text into discrete units called tokens. Sentence splitting divides unstructured documents into individual grammatical propositions using punctuation boundary rules.

Stemming is the heuristic process of stripping affixes to reduce words to their base root form. The Porter Stemmer is widely used in information retrieval, although it frequently produces non-words such as study becoming studi. In contrast, lemmatization is defined as the morphological analysis that transforms words to their dictionary canonical form called a lemma. Lemmatization requires part-of-speech context and vocabulary knowledge to accurately map running to run and better to good.

3. Statistical Feature Extraction & Vectorization

Term Frequency-Inverse Document Frequency is defined as a statistical numerical measure that evaluates how important a word is to a document within a collection. Term frequency measures the raw frequency of a word in a specific document. Inverse document frequency diminishes the weight of common terms that appear across all documents, penalizing ubiquitous stopwords like the, is, and at.

Luhn summarization is an extractive algorithm developed by Hans Peter Luhn at IBM in 1958. The Luhn frequency method scores sentences by calculating the clustering density of significant content words. Extractive summarization guarantees complete factual preservation because selected sentences are quoted verbatim from the source notes without artificial hallucination.

4. Linguistic Analytics & Sequence Modeling

Part-of-Speech tagging is defined as the grammatical annotation of words with their syntactic categories such as nouns, verbs, adjectives, and adverbs. The Averaged Perceptron tagger utilizes preceding word features and lexical patterns to achieve high tagging accuracy. Named Entity Recognition identifies real-world entities in

unstructured text, classifying terms into categories including PERSON, ORGANIZATION, GPE, and DATE.

In 2017, researchers at Google in California published the Transformer architecture based entirely on self-attention mechanisms. Readability formulas like the Flesch Reading Ease and Flesch-Kincaid Grade Level measure textual accessibility through syllable counts and sentence lengths.

Core Exam Takeaways:

- **Extractive Summarization** preserves 100% of the author original phrasing by selecting top-ranked sentences.
- **Lemmatization** produces legitimate dictionary headwords, whereas **Stemming** relies on rule-based suffix trimming.
- **TF-IDF** elevates domain-specific terminology while suppressing ubiquitous function words.
- **Hans Peter Luhn** introduced the normalized frequency clustering approach to automated document abstraction.